

Dharmendra Reddy Chitte

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SUMMARY

AI Engineer and M.S. Computer Science graduate (GPA: 4.0/4.0) with experience building **AI-powered systems and backend infrastructure** for real-world applications. Skilled in developing **LLM-driven agents, retrieval-augmented generation (RAG) pipelines, and scalable APIs** using Python, LangChain, and cloud-native tools. Experienced in designing data pipelines, integrating AI into production systems, and deploying services using Docker and AWS. Published research on **RAG-based intelligent systems in IEEE and Elsevier**, and passionate about building reliable AI platforms that solve real enterprise problems at scale.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, HTML, CSS

Data Engineering: Data Pipelines, ETL, Data Processing, Data Modeling, Data Ingestion, SQL Optimization

AI / Machine Learning: Machine Learning, NLP, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, Prompt Engineering, Embeddings, Semantic Search, TensorFlow, PyTorch, Hugging Face

Frameworks: LangChain, LangGraph, Django, Spring Boot, React, FastAPI, RAGAS

Cloud & Distributed Systems: AWS (ECS, ECR), Docker, REST APIs, Microservices, CI/CD, Linux

Databases: PostgreSQL, MySQL, Oracle, MongoDB, ChromaDB

Developer Tools: Git, Linux, MCP Server

EXPERIENCE

AI Engineer

Jan 2024 – Present

InfraTie Solutions

Stillwater, OK

- Designed and deployed an **LLM-based natural language interface** for querying large-scale infrastructure datasets, reducing information retrieval time by **60%**.
- Built a Python-based **NLP and RAG pipeline** that converts natural language questions into structured SQL queries, achieving **85%+ accuracy** using schema-aware retrieval.
- Applied **prompt engineering, embeddings, and semantic retrieval** to improve relevance and robustness across complex multi-table queries.
- Integrated the AI system into a Django application and deployed it using **Docker and AWS ECS** for scalable inference.
- Co-authored *Automated SQL Query Generation for an Intelligent Sewer Management System Using Large Language Models*, published at **IEEE CASE 2025**.
- Wrote **unit and integration tests** to validate system behavior and improve software reliability.

Software Developer

Jul 2022 – Aug 2023

Oracle Financial Services Software

Karnataka, India

- Optimized SQL queries and database interactions across core banking systems, improving transaction throughput by **35%** and reducing latency in high-traffic workflows.
- Developed backend microservices using **Spring Boot** and **Java**, delivering secure, scalable services across multiple client deployments.
- Built responsive UI components with JavaScript/HTML/CSS and collaborated with QA/product teams in **Agile** sprints to ship features reliably.
- Used Git-based workflows and CI/CD pipelines to deploy services and support production releases with minimal downtime.

Junior Web Developer Intern

Jan 2022 – Jun 2022

Cognizant

Hyderabad, India

- Developed web applications using **React**, JavaScript, and .NET, improving user experience and automating enterprise workflows.
- Designed and tested **REST APIs** to integrate front-end features with backend systems, focusing on reliability and maintainability.
- Worked with business stakeholders to translate requirements into technical designs and deliver features in an Agile environment.

EDUCATION

Oklahoma State University GPA: 4.0/4.0

Stillwater, OK

Master of Science in Computer Science

Aug 2023 – Aug 2025

- Coursework: Machine Learning, Natural Language Processing, Data Structures & Algorithms, Distributed Systems, Cloud Computing.

PROJECTS

AgenticRAG — Agentic Multi-Document RAG Assistant | *Python, LangChain, OpenAI, ChromaDB* | [GitHub](#)2026

- Built an **agentic RAG system** over a corpus of arXiv cs.AI research papers where the agent autonomously decides at each turn whether to **retrieve, ask a clarifying question, call an external tool, refuse, or answer**, instead of following a fixed retrieve-then-answer pipeline.
- Engineered an end-to-end ingestion pipeline for technical PDFs with **metadata-aware chunking, OpenAI embeddings, and persistent ChromaDB vector storage** for semantic search.
- Implemented retrieval beyond naive top-k using **query rewriting and cross-encoder reranking**, and designed layered **conversation, semantic, and episodic memory** so prior turns meaningfully shape retrieval and answers.
- Developed an **evaluation harness** scoring agent behavior on hand-written questions, including refusal and clarification cases, and added **observability** with per-query decision traces and logs.

Autonomous Coding Assistant — Agentic Software Engineering Tool | *LLMs, RAG, MCP, VectorDB* | [GitHub](#)2026

- Architected an **autonomous coding assistant** from scratch, inspired by agentic coding paradigms, capable of understanding, modifying, and generating code across entire repositories.
- Implemented a modular **agentic architecture** with planning, execution, tool-calling, memory, and repository reasoning components for autonomous software engineering workflows.
- Built intelligent code search, repository indexing, dependency analysis, and context management for project-wide code understanding and accurate code generation.
- Integrated filesystem tools, terminal execution, and iterative reasoning loops to automate debugging, refactoring, feature implementation, and documentation generation.

Healthcare RAG Retrieval Optimization | *LLMs, RAG, Information Retrieval*

2026

- Developed a healthcare-specific retrieval pipeline by fine-tuning **E5-Large-v2** embeddings and integrating **ColBERTv2** re-ranking for semantic search.
- Generated domain-specific training data using **Llama 3.3-70B**, creating synthetic question–document pairs from processed healthcare PDFs.
- Built scalable preprocessing workflows for PDF extraction, data cleaning, chunking, and vectorization to support retrieval-augmented applications.
- Improved retrieval accuracy from **63% to 77%** and benchmarked performance using the **BEIR** evaluation framework.

PUBLICATIONS

- **Automated SQL Query Generation for an Intelligent Sewer Management System Using Large Language Models** — IEEE CASE 2025. [Paper]
- **Domain-Specific SQL Generation with LLMs: A Hybrid Framework Combining Knowledge Graphs and Retrieval-Augmentation** — Elsevier. [Paper]
- **Human Action Recognition Based on Depth Maps, Skeleton and Sensor Images Using Deep Learning** — IEEE. [Paper]